



Morteza Zakeri

Curriculum Vitæ

"Towards better software systems by automating laborious engineering activities with Software II and Compiler II."

Education

- 2018—2023 **Ph.D. in Computer Engineering (Software)**, *Iran University of Science and Technology (IUST)*, Tehran, Iran
- Dissertation: "*Measuring and improving testability of software systems artifacts*"
 - Supervisor: Prof. [Saeed Parsa](#)
 - Advisor: Dr. [Mehrdad Ashtiani](#)
 - GPA: *19.11 out of 20* (Ranked 1st).
- 2016—2018 **M.Sc. in Computer Engineering (Software)**, *Iran University of Science and Technology (IUST)*, Tehran, Iran
- Thesis: "*Automatic test data generation in file format fuzzers*"
 - Supervisor: Prof. [Saeed Parsa](#)
 - GPA: *18.54 out of 20* (Ranked 1st)
- 2011—2015 **B.Sc. in Computer Engineering (Software)**, *Arak University*, Markazi, Iran
- Project: "*Designing and implementing a multi-agent system to participate in the multi-agent programming contest (MAPC'15)*"
 - Supervisor: Dr. [Vahid Rafe](#)
 - GPA: *18.18 out of 20* (Ranked 2nd)
- 2010—2011 **Pre-college in Mathematics Science**, *Beheshti Pre-college*, Isfahan, Iran
- GPA: *19.35 out of 20*
- 2008—2010 **Diploma in Mathematics and Physics**, *Beheshti High School (2nd and 3rd years) and Ibn-e-Sina High School (1st year)*, Isfahan, Iran
- GPA: *19.77 out of 20*

Research Interests

Software Engineering & *Machine Intelligence* Compiler systems, program analysis and transformation, software refactoring, testing, debugging, and repair; Automated and empirical software engineering, requirements engineering, and software quality assurance; Machine learning, deep learning, and natural language processing (NLP) for software engineering (AI4SE); Artificial intelligence applications in material (ceramics, metallurgy), civil, and biomedical engineering; Artificial intelligence applications in social and financial systems for entrepreneurship and management.

Honors and Awards

- Graduate study
- Awarded as an outstanding Ph.D. researcher, [IUST Exceptional Talents Office](#), Winter 2022 and Winter 2023.
 - Ranked 1st among all 60 students during the IUST M.Sc. program, Fall 2018
 - Ph.D. admission without entrance exam, IUST, Fall 2018.
 - Awarded as an outstanding M.Sc. graduate by [IUST Exceptional Talents Office](#), Winter 2018.
 - Awarded as an outstanding student by [IUST Exceptional Talents Office](#), Winter 2017.
- Undergraduate study
- Ranked 2nd among 31 students during my B.Sc. program at [Arak University](#).
 - Awarded as an outstanding student by the [Arak University Exceptional Talents Office](#), Fall 2014.
 - Ranked 1st in [Arak University](#) futsal tournaments, Winter 2014.
 - Awarded as *ethics* team in [Arak University](#) pantomime competitions, Winter 2014.

- High school
- Ranked 3rd in Isfahan physics laboratory contests, Spring 2010.
 - Ranked 2nd and 3rd in Isfahan regional students' chess tournaments, Fall 2008 and Fall 2010.

Skills

Theoretical computer science background

- Selected university courses with grades (my grade / total)
- Software engineering (19.5/20), software architectures (18.80/20), and object-oriented design (19.25/20)
 - Compiler design (20/20), advanced compilers (18/20), dependable software systems (18.80/20)
 - Formal languages and automata (19.25/20), algorithms design (18.50/20), and game theory (18/20)
 - Computer architectures (20/20), operating systems (20/20), distributed systems (19/20), and internet of things (20/20)
 - Database systems (20/20), data mining (19/20), graph/ network mining—complex dynamic networks (19.75/20), and text mining—natural language processing (20/20).

Applied computer science background

- Programming and markup languages
- Python and Java [expert], C, C++, and C# [proficient], Scratch [familiar]
 - Assembly (x86) and shell scripting [familiar]
 - XML, (X)HTML, CSS, YAML, JSON, DOM, and AJAX [familiar]

- Tools, frameworks, libraries, and IDEs
- **Compilers:** ANTLR [expert], LLVM, Roslyn, and JDT [familiar]
 - **Program analysis:** Understand, PMD, Doxygen, SonarQube, SourceMeter, WinDbg, and IDA Pro
 - **Program testing:** EvoSuite, JDART, AFL, Postman, JMeter, Acunetix, and Selenium
 - **Data science:** Scikit-learn, Tensorflow2 (Keras), NetworkX, SciPy, Pymoo, MLFlow, and Weka
 - **Visualization:** Seaborn, Graphviz, Cytoscape, and Matplotlib
 - **Application software development:** PyQt, JavaFX, Django, and Jason (multi-agent programming)
 - **Website building:** Wordpress, Joomla, Moodle, MkDocs, and Pelican
 - **Databases:** MySQL, Microsoft SQL-Server, and OrientDB (NoSQL)
 - **Dependency and code management tools:** Pip, Maven, and Git
 - **IDEs:** PyCharm, IntelliJ IDEA, Eclipse, Netbeans, VS Code, and Visual Studio.

- Software development methodologies, modeling, and management
- **Methodologies/Processes:** Agile: Specialized in *Scrum*— with test-driven development (TDD) and behavior-driven development (BDD) practices, rational unified process (RUP), and Oracle custom development method (CDM)
 - **Modeling languages:** Unified modeling language (UML), business process model and notation (BPMN), entity relationship diagram (ERD), Markov chain, and Petri net
 - **Modeling tools:** Enterprise Architect, Visual Paradigm, and Visio
 - **Project management and communication:** Scrum (Jira, Trello), Confluence, Slack.

- General computer skills
- **Operating systems:** Microsoft Windows and Linux (Ubuntu desktop/server, and Kali)
 - **Virtualization:** VMware Workstation and ESXi
 - **Typesetting and presentation:** Microsoft Office and L^AT_EX
 - **Multi-media:** Camtasia.

Other

- Languages **Persian:** Native, **English:** Good (English degree: MSRT exam with score of 74/100), and **Arabic:** Basic.
- Sport and hobbies Futsal, hiking, chess, board games, and watching (scientific videos and documentary films).

Publications

Selected journal papers (JCR)

- [JCR-025] Narges Abbasi, Saeed Parsa, and Morteza Zakeri. **A Systematic Literature Review on Design for Testability Approaches in Software Systems**. In: *ACM Comput. Surv.* 58.12 (May 2026). ISSN: 0360-0300. DOI: 10.1145/3787493.
- [JCR-024] Elham Azhir, Morteza Zakeri, Yasaman Abedini, and Mojtaba Mostafavi Ghahfarokhi. **QualCode: A data-driven framework for predicting software maintainability based on ISO/IEC 25010**. In: *Science of Computer Programming* 250 (2026), p. 103399. ISSN: 0167-6423. DOI: <https://doi.org/10.1016/j.scico.2025.103399>.
- [JCR-023] Alireza Khanlari, Ali Reza Eivani, Morteza Zakeri, Jie Zhou, Hamid Reza Jafarian, and Morteza Tayebi.

- Machine learning-assisted prediction of the hardness of additively manufactured and heat-treated Ti-6Al-4V alloy.** In: *Results in Engineering* 30 (2026), p. 110727. ISSN: 2590-1230. DOI: <https://doi.org/10.1016/j.rineng.2026.110727>.
- [JCR-022] Morteza Zakeri, Fatemeh Abdi, and Fatemeh Bagheri. **Enhancing software quality attributes through multi-dimensional refactoring at source-level.** In: *Science of Computer Programming* 251 (2026), p. 103434. ISSN: 0167-6423. DOI: <https://doi.org/10.1016/j.scico.2025.103434>.
- [JCR-021] Rasoul Rezvani-Jalal, Morteza Zakeri, Saeed Parsa, and Amin Hasan-Zarei. **Enhancing malware detection reliability in non-executable files using confidence score prediction.** Aug. 2026. DOI: [10.1007/s10207-026-01312-3](https://doi.org/10.1007/s10207-026-01312-3).
- [JCR-020] Fateme Bagheri-Galle, Saeed Parsa, and Morteza Zakeri. **A systematic literature review on transformation for testability techniques in software systems.** In: *Information and Software Technology* (2025), p. 107788. DOI: [10.1016/j.infsof.2025.107788](https://doi.org/10.1016/j.infsof.2025.107788).
- [JCR-019] Arash Ghasemi, Ali Reza Eivani, Omid Banapour Ghaffari, Morteza Zakeri-Nasrabadi, Manojit Ghosh, Seyed Mahmood Fatemi, and Hamid Reza Jafarian. **Machine learning for predicting high-temperature strength of austenitic stainless steels based on composition and deformation parameters.** In: *Journal of Materials Science* 60.47 (2025), pp. 24665–24683. DOI: <https://doi.org/10.1007/s10853-025-11805-6>.
- [JCR-018] Saeed Parsa, Morteza Zakeri-Nasrabadi, and Burak Turhan. **Testability-driven development: an improvement to the TDD efficiency.** In: *Computer Standards & Interfaces* 91 (Jan. 2025), p. 103877. ISSN: 09205489. DOI: [10.1016/j.csi.2024.103877](https://doi.org/10.1016/j.csi.2024.103877).
- [JCR-017] Alireza Ardalani, Saeed Parsa, Morteza Zakeri-Nasrabadi, and Alexander Chatzigeorgiou. **Supporting single responsibility through automated extract method refactoring.** In: *Empirical Software Engineering* 29 (1 Jan. 2024), p. 28. ISSN: 1382-3256. DOI: [10.1007/s10664-023-10427-3](https://doi.org/10.1007/s10664-023-10427-3).
- [JCR-016] Omid Banapour Ghaffari, Bijan Eftekhari Yekta, and Morteza Zakeri-Nasrabadi. **Designing high-performance ion-exchangeable glasses with multi-objective optimization and machine learning.** In: *Ceramics International* (Aug. 2024). ISSN: 0272-8842. DOI: <https://doi.org/10.1016/j.ceramint.2024.08.141>.
- [JCR-015] Omid Banapour Ghaffari, Bijan Eftekhari Yekta, and Morteza Zakeri-Nasrabadi. **Estimating “depth of layer” (DOL) in ion-exchanged glasses using explainable machine learning.** In: *Materialia* (Jan. 2024), p. 102027. ISSN: 25891529. DOI: [10.1016/j.mtla.2024.102027](https://doi.org/10.1016/j.mtla.2024.102027).
- [JCR-014] Roshan Golmohammadi, Saeed Parsa, and Morteza Zakeri-Nasrabadi. **Dynamic domain testing with multi-agent Markov chain Monte Carlo method.** In: *Soft Computing* (2024). DOI: [10.1007/s00500-024-09680-5](https://doi.org/10.1007/s00500-024-09680-5).
- [JCR-013] Ali Majidzadeh, Mehrdad Ashtiani, and Morteza Zakeri-Nasrabadi. **Multi-type requirements traceability prediction by code data augmentation and fine-tuning MS-CodeBERT.** In: *Computer Standards & Interfaces* 90 (Aug. 2024), p. 103850. ISSN: 09205489. DOI: [10.1016/j.csi.2024.103850](https://doi.org/10.1016/j.csi.2024.103850).
- [JCR-012] Mostafa Mir, Farnad Nasirzadeh, Morteza Zakeri, Aron T. Hill, and Chandan Karmakar. **Assessing neural markers of attention during exposure to construction noise using machine learning classification of electroencephalogram data.** In: *Building and Environment* 261 (Aug. 2024), p. 111754. ISSN: 03601323. DOI: [10.1016/j.buildenv.2024.111754](https://doi.org/10.1016/j.buildenv.2024.111754).
- [JCR-011] Morteza Zakeri-Nasrabadi and Saeed Parsa. **Natural language requirements testability measurement based on requirement smells.** In: *Neural Computing and Applications* (Apr. 2024). ISSN: 0941-0643. DOI: [10.1007/s00521-024-09730-x](https://doi.org/10.1007/s00521-024-09730-x).
- [JCR-010] Morteza Zakeri-Nasrabadi, Saeed Parsa, and Sadegh Jafari. **Measuring and improving software testability at the design level.** In: *Information and Software Technology* (Oct. 2024), p. 107511. ISSN: 09505849. DOI: [10.1016/j.infsof.2024.107511](https://doi.org/10.1016/j.infsof.2024.107511).
- [JCR-009] Eisa Zarepour, Mohammad Reza Mohammadi, Morteza Zakeri-Nasrabadi, Sara Aein, Razieh Sangsari, Leila Taheri, Mojtaba Akbari, and Ali Zabihallahpour. **BiliBin: An intelligent mobile phone-based platform to monitor newborn jaundice.** In: *IRANIAN JOURNAL OF ELECTRICAL AND ELECTRONIC ENGINEERING* 20.3 (Sept. 2024). DOI: [10.22068/IJEEE.20.3.3374](https://doi.org/10.22068/IJEEE.20.3.3374).
- [JCR-008] Morteza Zakeri-Nasrabadi, Saeed Parsa, Masoud Ekhtiarzadeh, Chanchal Roy, and Mohammad Ramezani. **A systematic literature review on source code similarity measurement: techniques, applications, and challenges.** In: *Journal of Systems and Software* (2023). DOI: [10.1016/j.jss.2023.111796](https://doi.org/10.1016/j.jss.2023.111796).

- [JCR-007] Morteza Zakeri-Nasrabadi, Saeed Parsa, Ehsan Esmaili, and Fabio Palomba. **A systematic literature review on the code smells datasets and validation mechanisms**. In: *ACM Computing Surveys* (May 2023). ISSN: 0360-0300. DOI: [10.1145/3596908](https://doi.org/10.1145/3596908).
- [JCR-006] Saeed Parsa, Morteza Zakeri-Nasrabadi, Masoud Ekhtiarzadeh, and Mohammad Ramezani. **Method name recommendation based on source code metrics**. In: *Journal of Computer Languages* 74 (2023), p. 101177. ISSN: 2590-1184. DOI: <https://doi.org/10.1016/j.col.2022.101177>.
- [JCR-005] Morteza Zakeri-Nasrabadi and Saeed Parsa. **Learning to predict test effectiveness**. In: *International Journal of Intelligent Systems* 37.8 (2022), pp. 4363–4392. DOI: <https://doi.org/10.1002/int.22722>.
- [JCR-004] Morteza Zakeri-Nasrabadi and Saeed Parsa. **An ensemble meta-estimator to predict source code testability**. In: *Applied Soft Computing* 129 (2022), p. 109562. ISSN: 1568-4946. DOI: <https://doi.org/10.1016/j.asoc.2022.109562>.
- [JCR-003] Mahnoosh Shahidi, Mehrdad Ashtiani, and Morteza Zakeri-Nasrabadi. **An automated extract method refactoring approach to correct the long method code smell**. In: *Journal of Systems and Software* 187 (May 2022), p. 111221. ISSN: 0164-1212. DOI: <https://doi.org/10.1016/j.jss.2022.111221>.
- [JCR-002] Morteza Zakeri-Nasrabadi, Hamideh Tabibi, Mahsa Salmani, Mahdiah Torkashvand, and Eisa Zarepour. **A comprehensive survey on non-invasive wearable bladder volume monitoring systems**. In: *Medical & Biological Engineering & Computing* 59.7-8 (Aug. 2021), pp. 1373–1402. ISSN: 0140-0118. DOI: [10.1007/s11517-021-02395-x](https://doi.org/10.1007/s11517-021-02395-x).
- [JCR-001] Morteza Zakeri-Nasrabadi, Saeed Parsa, and Akram Kalaei. **Format-aware learn&fuzz: deep test data generation for efficient fuzzing**. In: *Neural Computing and Applications* (June 2020). ISSN: 0941-0643. DOI: [10.1007/s00521-020-05039-7](https://doi.org/10.1007/s00521-020-05039-7).

Selected conference papers

- [Conf-004] Soroush Hashemifar and Morteza Zakeri-Nasrabadi. **Deep identification of plant diseases**. In: *2024 20th CSI International Symposium on Artificial Intelligence and Signal Processing (AISP)*. Tehran: IEEE, Feb. 2024, pp. 1–6. ISBN: 979-8-3503-8394-2. DOI: [10.1109/AISP61396.2024.10475267](https://doi.org/10.1109/AISP61396.2024.10475267).
- [Conf-003] Ali Majidzadeh, Mehrdad Ashtiani, and Morteza Zakeri-Nasrabadi. **Code data augmentation to improve language model's performance in requirement to code traceability link recovery**. In: *Proceedings of the 9th International Conference on Web Research*. Tehran, Iran: University of Science and Culture (In Persian), 2023.
- [Conf-002] Morteza Zakeri-Nasrabadi and Saeed Parsa. **Learning to predict software testability**. In: *Proceedings of the 26th International Computer Conference, Computer Society of Iran (CSICC)*. Tehran: IEEE, Mar. 2021, pp. 1–5. ISBN: 978-1-6654-1241-4. DOI: [10.1109/CSICC52343.2021.9420548](https://doi.org/10.1109/CSICC52343.2021.9420548).
- [Conf-001] Zahra Zakeri-Nasrabadi and Morteza Zakeri-Nasrabadi. **Analysis social phenomena using machine learning techniques: a mixed research framework**. In: *Proceedings of the first conference on artificial intelligence and soft computing in humanities (AISCH-2019)*. Tehran, Iran: Allameh Tabataba'i University (In Persian), 2019, pp. 120–127.

Selected journal papers (non-JCR)

- [ISC-003] Morteza Zakeri and Saeed Parsa. **Learning to predict software testability axiomatically**. In: *The CSI Journal on Computer Science and Engineering* 19.1 (2025), pp. 43–53. ISSN: 1735-9031.
- [ISC-002] Morteza Zakeri-Nasrabadi, Saeed Parsa, and Zahra Hayati. **Automatic test data generation to improve fault-localization based on causal-statistical analysis**. in Persian. In: *Journal of Soft Computing and Information Technology* 12 (2024), pp. 1–11. ISSN: 2383-1006.
- [ISC-001] Morteza Zakeri-Nasrabadi and Saeed Parsa. **Automatic test data generation in file format fuzzers**. in Persian. In: *Electronic and Cyber Defense* 8.1 (2020), pp. 1–16. ISSN: 2322-4347.

Archived and under review papers

- [Preprint-003] Soroush Hashemifar, Saeed Parsa, and Morteza Zakeri-Nasrabadi. **Mitigating backdoors within deep neural networks in data-limited configuration**. 2024. DOI: <https://doi.org/10.48550/arXiv.2311.07417>.
- [Preprint-002] Morteza Zakeri-Nasrabadi, Saeed Parsa, and Mohamed Wiem Mkaouer. **Flipped boosting of automatic test data generation frameworks through a many-objective program transformation approach**. 2023. DOI: <http://dx.doi.org/10.2139/ssrn.4373904>.
- [Preprint-001] Saeed Parsa, Morteza Zakeri-Nasrabadi, and Ehsan Esmaili. **Predicting code quality attributes based on code smells**. 2023.

Theses

- [Thesis-003] Morteza Zakeri-Nasrabadi. **Measuring and improving testability of software systems artifacts**. in Persian. Ph.D. dissertation. Iran University of Science and Technology (IUST), School of Computer Engineering, Sept. 2022.
- [Thesis-002] Morteza Zakeri-Nasrabadi. **Automatic test data generation in file format fuzzers**. in Persian. M.Sc. thesis. Iran University of Science and Technology (IUST), School of Computer Engineering, Sept. 2018.
- [Thesis-001] Morteza Zakeri-Nasrabadi. **Designing and implementing a multi-agent system to participate in the multi-agent programming contest (MAPC'15)**. in Persian. B.Sc. project. Arak University, Faculty of Engineering, Sept. 2015.

Complete list of publications

- Research profiles The *up-to-date* list of my publications are available in the following research profiles:
- Web of Science™ (Researcher ID: Y-6393-2018): <http://www.researcherid.com/rid/Y-6393-2018>
 - Scopus (Scopus Author ID: 57219747851): <https://www.scopus.com/authid/detail.uri?authorId=57219747851>
 - ORCID (0000-0003-4289-0606): <https://orcid.org/0000-0003-4289-0606>
 - Google scholar: <https://scholar.google.com/citations?user=km5DzwwAAAAJ&hl=en>
 - ResearchGate: <https://www.researchgate.net/profile/Morteza-Zakeri>
 - DBLP: <https://dblp.org/pid/232/3298.html>
 - ACM: <https://dl.acm.org/profile/99660089625>

Professional Activities

Academic and industry experiences

- 2026—Now **ICT deputy**, Department of Computer Engineering (<https://ce.aut.ac.ir/>), Amirkabir University of Technology (AUT), Main Campus, Tehran, Iran
- Managing and maintaining the department's technology infrastructure and data center and ICT services, policies, and plans
- 2025—Now **Assistant professor**, Department of Computer Engineering (<https://ce.aut.ac.ir/>), Amirkabir University of Technology (AUT), Main Campus, Tehran, Iran
- Working as a faculty member (<https://aut.ac.ir/cv/26642/Morteza-Zakeri>)
 - Director of Intelligent Software Engineering research laboratory (<https://www.m-zakeri.ir/lab/en/>)
- 2024—2025 **Postdoctoral researcher**, School of Computer Science (<https://cs.ipm.ac.ir/>), Institute for Research in Fundamental Sciences (IPM), Farmanieh Campus, Tehran, Iran
- Working on software design testability, inverse design, and intelligent systems design
 - <https://cs.ipm.ac.ir/PostdocResearchers.aspx>
- 2023—2024 **University lecturer (adjunct professor)**, Computer science and engineering courses, Iran University of Science and Technology (IUST), University of Tehran, K. N. Toosi University of Technology (KNTU), Amirkabir University of Technology (AUT), and Sharif University of Technology (SUT), Tehran, Iran
- 2018—2023 **Ph.D. student and candidate**, Reverse Engineering Research Laboratory (<http://reverse.iust.ac.ir/>), Iran University of Science and Technology, Tehran, Iran
- Developing an automated program analysis library, OpenUnderstand (<https://m-zakeri.github.io/OpenUnderstand>)
 - Developing an automated refactoring engine, CodART (<https://m-zakeri.github.io/CodART>)
 - Developing software requirements quality measurement tool, ARTA (<https://m-zakeri.github.io/ARTA>)
 - Developing source code testability measurement tool, ADAFEST (<https://m-zakeri.github.io/ADAFEST>)
 - Developing a file format fuzzer, DeepFuzz (<https://m-zakeri.github.io/iust-deep-fuzz>).
 - Supervisor: Prof. Saeed Parsa (<http://parsa.iust.ac.ir>)
- 2021—2022 **Project manager and machine learning engineer**, Fanavaran Denshgar Co. (<https://www.dantech.ir/>), Tehran, Iran
- Intelligent anti-money laundering (AML) system
 - Big data analysis on banking data
- 2020—2021 **Research assistant**, Automated Software Engineering Laboratory (<http://ase.ce.sharif.ir/>), Sharif University of Technology, Tehran, Iran
- Designing and implementing a software maintainability measurement tool, QualCode (<https://qualcode.ir/>)
 - Project supervisor: Dr. Abbas Heydarnoori (<http://sharif.edu/~heydarnoori>)
 - Funded by Iran's National Elites Foundation and MCI R&D Center
- 2019—2020 **Research assistant**, Iranian Online Smart Monitoring (Riz-Payesh) Healthcare Company, Tehran, Iran
- Designing a wearable bladder monitoring system (WBMS)
 - Project supervisor: Dr. Eisa Zarepour (<http://webpages.iust.ac.ir/zarepour>)
 - Funded by Iran's National Elites Foundation
- 2015—2016 **Software engineer**, Pars Sina Azeen Consulting Engineers Company (Parsina), Khorramabad, Lorestan
- Designing and developing Parsina bridge management system (PBMS)

- Mar—Aug, 2015 **Intern**, *University Data Center and Computer Engineering Laboratories, Arak University, Arak, Markazi*
 ○ Building AVR and ARM micro-controllers educational boards, launching the faculty cloud-center based on 2X OS, and rewriting and revising laboratories pamphlets and handbooks

Teaching experiences

- Spring 2026 **Lecturer (Advanced software testing and program analysis —graduate)**, *Amirkabir University of Technology (Tehran Polytechnique)*, Tehran, Iran
 ○ Course webpage: <https://www.m-zakeri.ir/SoftwareTesting/>
- Spring 2026 **Lecturer (Software engineering I —undergraduate)**, *Amirkabir University of Technology (Tehran Polytechnique)*, Tehran, Iran
 ○ Course webpage: <https://www.m-zakeri.ir/SE1/>
- Spring 2026 **Lecturer (Research and presentation methods —undergraduate)**, *Amirkabir University of Technology (Tehran Polytechnique)*, Tehran, Iran
 ○ Course webpage: <https://www.m-zakeri.ir/lab/courses>
- Fall 2025 **Lecturer (Principles and patterns in software engineering —graduate)**, *Amirkabir University of Technology (Tehran Polytechnique)*, Tehran, Iran
 ○ Course webpage: <https://www.m-zakeri.ir/SEP/>
- Fall 2025 **Lecturer (Entrepreneurship—undergraduate)**, *Amirkabir University of Technology (Tehran Polytechnique)*, Tehran, Iran
 ○ Course webpage: <https://www.m-zakeri.ir/Entrep>
- Fall 2025 **Lecturer (Theory of formal languages and automata—undergraduate)**, *Amirkabir University of Technology (Tehran Polytechnique)*, Tehran, Iran
 ○ Course webpage: <https://www.m-zakeri.ir/lab/courses>
- Fall 2024—Spring 2025 **Lecturer (Compiler design—undergraduate)**, *Amirkabir University of Technology (Tehran Polytechnique)*, Tehran, Iran
 ○ Course webpage: <https://m-zakeri.github.io/Compilers>
- Spring 2024—Spring 2025 **Lecturer (Fundamentals of computer and programming—undergraduate)**, *Amirkabir University of Technology (Tehran Polytechnique)*, Tehran, Iran
 ○ Course webpage: <https://m-zakeri.github.io/CP>
- Fall 2024 **Lecturer (Compiler design—undergraduate)**, *Sharif University of Technology*, Tehran, Iran
 ○ Course webpage: <https://m-zakeri.github.io/Compilers>
- Spring 2023 **Lecturer (Object-oriented systems design —undergraduate)**, *K. N. Toosi University of Technology*, Tehran, Iran
- Spring 2023 **Lecturer (Software Testing—undergraduate)**, *University of Tehran (Fouman Faculty of Engineering)*, Gilan, Iran
 ○ Course webpage: <https://m-zakeri.github.io>
- Spring 2023 **Lecturer (Algorithm design—undergraduate)**, *University of Tehran (Fouman Faculty of Engineering)*, Gilan, Iran
 ○ Course webpage: <https://m-zakeri.github.io/Algorithms>
- Fall 2023 **Lecturer (Special topics in software engineering—graduate)**, *University of Tehran (Kish International Campus)*, Tehran, Iran
- Fall 2023 **Lecturer (Advanced software engineering—graduate)**, *University of Tehran (Kish International Campus)*, Tehran, Iran
- Fall 2023 **Lecturer (Artificial intelligence—undergraduate)**, *K. N. Toosi University of Technology*, Tehran, Iran
 ○ Course webpage: <https://m-zakeri.github.io/AI>
- Fall 2023 **Lecturer (Database systems design—undergraduate)**, *K. N. Toosi University of Technology*, Tehran, Iran
 ○ Course webpage: <https://m-zakeri.github.io/DatabaseDesign/>
- Fall 2023 **Lecturer (Programming languages and compiler design—undergraduate)**, *University of Tehran (Fouman Faculty of Engineering)*, Gilan, Iran
 ○ Course webpage: <https://m-zakeri.github.io/Compilers>
- Spring 2023 **Lecturer (Fundamentals of compiler design—undergraduate)**, *Iran University of Science and Technology*, Tehran, Iran
 ○ Course webpage: <https://m-zakeri.github.io/IUSTCompiler>

- Fall 2022 **Teaching assistant (Software architectures—graduate)**, *Iran University of Science and Technology*, Tehran, Iran
- Instructor(s): [Dr. Mehrdad Ashtiani](#)
 - Responsibilities: Designing and grading assignments and projects, holding extra office hours, and students' seminars
- 2017—2022 **Teaching assistant (Fundamental of compiler design—undergraduate)**, *Iran University of Science and Technology*, Tehran, Iran
- Instructor(s): [Dr. Saeed Parsa](#)
 - Web-page: <http://parsa.iust.ac.ir/courses/compilers>
 - Responsibilities: Designing and grading assignments, holding extra office hours, and editing lecture notes.
 - Funded by Iran's National Elites Foundation
- 2018—2022 **Teaching assistant (Advanced compiler—graduate)**, *Iran University of Science and Technology*, Tehran, Iran
- Instructor(s): [Dr. Saeed Parsa](#)
 - Web-page: <http://parsa.iust.ac.ir/courses/advanced-compilers/>
 - Responsibilities: Designing and grading assignments and projects, holding extra office hours, and editing lecture notes.
- 2019—2021 **Teaching assistant (Advanced software engineering—graduate)**, *Iran University of Science and Technology*, Tehran, Iran
- Instructor(s): [Dr. Saeed Parsa](#)
 - Web-page: <http://parsa.iust.ac.ir/courses/advanced-software-engineering/>
 - Responsibilities: Designing and grading assignments and projects, holding extra office hours, and students' seminars, and updating lecture notes.
- Spring 2020 **Teaching assistant (Game theory—graduate)**, *Iran University of Science and Technology*, Tehran, Iran
- Instructor(s): [Dr. Vesal Hakami](#)
 - Web-page: <https://m-zakeri.github.io/game-theory.html#game-theory>
 - Responsibilities: Designing and grading assignments and projects, holding TA classes.
- Spring 2020 **Teaching assistant (Complex dynamic networks—graduate)**, *Iran University of Science and Technology*, Tehran, Iran
- Instructor(s): [Dr. Hossein Rahmani](#)
 - Web-page: <https://m-zakeri.github.io/dynamic-complex-network.html#dynamic-complex-network>
 - Responsibilities: Designing and grading assignments and projects.
- More My teaching courses: <https://www.m-zakeri.ir/list-of-my-teaching-courses.html>

Student mentoring

- AUT section

- 2025—2027 **Amin Zibayi**, *M.Sc.*, Thesis area: AI-native software and agentic engineering, Current: Working on his thesis
- 2025—2027 **Parsa Barikbin**, *M.Sc.*, Thesis area: Fuzz testing and automatic exploit generation, Current: Working on his thesis
- 2025—2027 **Romina Firuzi**, *M.Sc.*, Thesis area: Fuzz testing and software supply chain attacks, Current: Working on her thesis
- 2024—2026 **Javad Besharati-far**, *M.Sc.*, Thesis area: software remodularization and evolutionary coupling, Current: Working on his thesis

- IUST section

- 2021—2023 **Ali Majidzadeh**, *M.Sc.*, Thesis area: software requirements traceability, Next: Software engineering in industry
- 2020—2022 **Alireza Ardalani**, *M.Sc.*, Thesis area: automated refactoring, goal modeling, Next: Ph.D. at New Jersey Institute of Technology (NJIT)
- 2020—2022 **Ehsan Esmaeili**, *M.Sc.*, Thesis area: code smell detection, software quality, Next: Software engineering in industry
- 2020—2022 **Masoud Ekhtiarzadeh**, *M.Sc.*, Thesis area: code recommendation, automated rename refactoring, Next: Software engineering in industry
- 2020—2022 **Mohammad Ramezani**, *M.Sc.*, Thesis area: code recommendation, software readability, Next: Software engineering in industry
- 2019—2021 **Mahnoosh Shahidi**, *M.Sc.*, Thesis area: automated refactoring, batch refactoring, Next: Software engineering in industry
- 2021—2022 **Sadegh Jafari**, *B.Sc.*, Thesis area: software testability, refactoring to design patterns, Next: M.Sc. at Iran University of Science and Technology (IUST)

- 2020—2021 **Ali Ayati**, B.Sc., Thesis area: automated refactoring, refactoring engines, Next: Ph.D. at Texas A&M University
- 2019—2020 **Mohsen Farahanchi**, B.Sc., Thesis area: software testing, test suite minimization, Next: M.Sc. at Shahid Beheshti University

Services

- 2025 **PC Member**, 6th National Informatics Conference (NIC'25), <https://cs.ipm.ac.ir/nic/1403>, Institute for Research in Fundamental Sciences (IPM), Feb. 26-27, 2025, Tehran, Iran.
- 2025 **Reviewer**, 30th Asia and South Pacific Design Automation Conference (ASP-DAC), <https://www.aspdac.com>, Tokyo Odaiba Miraikan, Japan
- 2024—Now **Journal reviewer**, Scientific Reports, <https://www.nature.com/srep>
- 2024—Now **Journal reviewer**, ACM Transactions on Software Engineering and Methodology, <https://dl.acm.org/journal/tosem>
- 2023—Now **Journal reviewer**, Neural Computing and Applications, <https://link.springer.com/journal/521>
- 2023—Now **Journal reviewer**, Science of Computer Programming, <https://www.sciencedirect.com/journal/science-of-computer-programming>
- 2023 **Journal reviewer**, Medical & Biological Engineering & Computing, <https://www.springer.com/journal/11517>
- 2022 **Journal reviewer**, The ISC International Journal of Information Security (ISecure), <http://www.isecure-journal.com>
- 2022 **Journal reviewer**, TELKOMNIKA (Telecommunication Computing Electronics and Control), <http://telkomnika.uad.ac.id/index.php/TELKOMNIKA>
- 2022 **Journal reviewer**, Artificial Intelligence Review, <https://www.springer.com/journal/10462>
- 2021 **Journal reviewer**, Communications in Combinatorics, Cryptography & Computer Science (CCCS), <http://vonneumann-publishing.com/cccs>
- 2020 **Reviewer**, 5th International Conference on Combinatorics, Cryptography, Computer Science, and Computing (I4C'20), <http://i4c.iust.ac.ir/index.php?lang=en>, Tehran, Iran
- 2019 **Reviewer**, 25th International Computer Conference, Computer Society of Iran (CSICC'20), <http://csicc2020.iust.ac.ir/en/reviewers-committee.html>, Tehran, Iran
- More Refer to my ORCID: <https://orcid.org/0000-0003-4289-0606>

Memberships

- 2023—Now Professional member, Association of Computing Machinery (ACM) <https://dl.acm.org/profile/99660089625>

Personal Info. and References

- ▷ Find more - More information, including my presentations, talks, teaching resources, and open-source projects can be found on my website: <https://m-zakeri.github.io>. Please avoid printing this resume to preserve clickable links and support environmental sustainability.
- Personal email: m-zakeri@live.com
 - GitHub profile: <https://github.com/m-zakeri>
 - Research Lab. website: <https://m-zakeri.github.io/lab/en>
 - PhD student webpage: http://webpages.iust.ac.ir/morteza_zakeri
- ▷ References
1. Prof. [Saeed Parsa](mailto:parsa@iust.ac.ir) - Iran University of Science and Technology (parsa@iust.ac.ir)
 2. Prof. [Mohammad Abdollahi Azgomi](mailto:azgomi@iust.ac.ir) - Iran University of Science and Technology (azgomi@iust.ac.ir)
 3. Prof. [Mohammadreza Razazi](mailto:razzazi@aut.ac.ir) - Amirkabir University of Technology (razzazi@aut.ac.ir)
 4. Dr. [Fabio Palomba](mailto:fpalomba@unisa.it) - University of Salerno (fpalomba@unisa.it)
 5. Dr. [Mehrdad Ashtiani](mailto:m_ashtiani@iust.ac.ir) - Iran University of Science and Technology (m_ashtiani@iust.ac.ir)
 6. Dr. [Abbas Heydarnoori](mailto:aheydar@bgsu.edu) - Bowling Green State University (aheydar@bgsu.edu)
 7. Dr. [Vahid Rafeh](mailto:v.rafe@gold.ac.uk) - Goldsmiths, University of London (v.rafe@gold.ac.uk)
- More references are available upon request.

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